


## Control and Cybernetics

Feedback control and cybernetic principles laid the foundation for self-regulating systems.

→ feedback




## Planning and World Models

Planning and world models let machines reason about future states and actions.

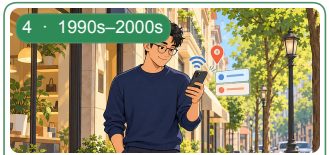
→ future states




## BDI and Agent Theory

Belief-desire-intention models formalized how agents reason and act toward goals.

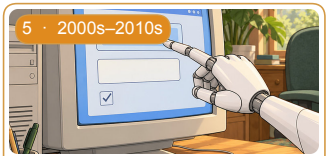
→ goals & intentions




## Ubiquitous and Context-Aware Computing

Computing embedded in everyday life: systems that sense context and adapt.

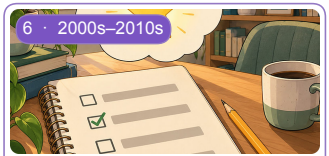
→ context




## Mixed-Initiative Interaction

User input combined with agent suggestions and actions for better decisions.

→ when to step in




## Goal Reasoning and Intrinsic Motivation

How agents generate goals, prioritize, and pursue what matters.

→ self-set goals




## Foundation Models and Tool-Using Agents

LLMs and tools enabled flexible reasoning and execution across real-world tasks.

→ all six




## Proactive AI agents

inherit all six capacities:

Awareness · Anticipation · Agenda · Arbitration · Action · Adaptation